

are central in the rest of this paper. It holds that

$$w_{\perp}^T \bar{A} w_{\perp} \geq \rho \Lambda w_{\perp}^T M w_{\perp} \quad \text{for all } w_{\perp} \in \bar{E}_{\leq \rho \Lambda}^{\perp} \quad (4.1)$$

Let $W_{\perp}$ be the basis matrix of $\bar{E}_{\leq \rho \Lambda}^{\perp}$. Note that W can be obtained by solving few lowest eigenmodes of the pencil $(\bar{A}, M)$, whereas $W_{\perp}$ is large, dense, and never available in a practical computation. In the following, we use the notation

$$B_{\perp} := W_{\perp}^T B W_{\perp}, \quad \forall B \in \mathbb{R}^{n \times n}. \quad (4.2)$$

We proceed to split eigenvectors of (1.1) as the sum of a component in $\bar{E}_{\leq \rho \Lambda}$ and a correction term in $\bar{E}_{\leq \rho \Lambda}^{\perp}$. A suitable splitting is given in Lemma 4.2 after a technical result.

Lemma 4.1. Assume that $A : S \rightarrow \mathbb{S}_{++}^{n \times n}$ is spectrally equivalent to $\bar{A}$ as in (3.1) with constants α and β . Let $\Lambda > 0$ and ρ be such that $\alpha \rho > 1$ and $\rho > 1$. Then the matrix $A_{\perp}(\sigma) - tM_{\perp}$ is positive definite for all $(\sigma, t) \in S \times (0; \Lambda)$.

Proof. For all $(\sigma, t) \in S \times (0; \Lambda)$ and $w_{\perp} \in \bar{E}_{\leq \rho \Lambda}^{\perp} \setminus \{0\}$, from the spectral equivalence of $A(\sigma)$ with $\bar{A}$, we get

$$w_{\perp}^T A(\sigma) w_{\perp} - t w_{\perp}^T M w_{\perp} \geq \alpha w_{\perp}^T \bar{A} w_{\perp} - t w_{\perp}^T M w_{\perp} \geq (\alpha \rho \Lambda - t) \|w_{\perp}\|_M^2 > 0 \quad (4.3)$$

as $w_{\perp} \in \bar{E}_{\leq \rho \Lambda}^{\perp}$. $\square$

Lemma 4.2. Make the same assumptions as in Lemma 4.1. Let $\sigma \in S$ and $(\lambda(\sigma), x(\sigma)) \in (0; \Lambda) \times \mathbb{R}^n \setminus \{0\}$ be a solution to (1.1). In addition, let $W_{\perp}$ be a basis matrix of $\bar{E}_{\leq \rho \Lambda}^{\perp}$. Then there holds that

$$x(\sigma) = \bar{x}(\sigma) + Z(\sigma, \lambda(\sigma)) \bar{x}(\sigma), \quad (4.4)$$

for $\bar{x}(\sigma) \in \bar{E}_{\leq \rho \Lambda}$ and correction operator $Z(\sigma, t) : S \times (0, \Lambda) \mapsto \mathbb{R}^{n \times n}$ defined as

$$Z(\sigma, t) = W_{\perp} (W_{\perp}^T (A(\sigma) - tM) W_{\perp})^{-1} W_{\perp}^T \delta A(\sigma) \quad (4.5)$$

with $\delta A(\sigma) = A(\sigma) - \bar{A}$.

Proof. Let $\sigma \in S$ and split $x(\sigma) = \bar{x}(\sigma) + r(\sigma)$ with $\bar{x}(\sigma) \in \bar{E}_{\leq \rho \Lambda}$ and $r(\sigma) = W_{\perp} \gamma(\sigma) \in \bar{E}_{\leq \rho \Lambda}^{\perp}$. Then as $(\lambda(\sigma), x(\sigma)) \in (0; \Lambda) \times \mathbb{R}^n$ is a solution of (1.1), we get

$$(A(\sigma) - \lambda(\sigma)M) \bar{x}(\sigma) = -(A(\sigma) - \lambda(\sigma)M) W_{\perp} \gamma(\sigma). \quad (4.6)$$

By orthogonality, we first observe that $W_{\perp}^T M \bar{x}(\sigma) = 0$ and $W_{\perp}^T \bar{A} \bar{x}(\sigma) = 0$. By multiplication with basis matrix $W_{\perp}^T$ and using these identities gives

$$W_{\perp}^T (A(\sigma) - \bar{A}) \bar{x}(\sigma) = -W_{\perp}^T (A(\sigma) - \lambda(\sigma)M) W_{\perp} \gamma(\sigma). \quad (4.7)$$

Then from Lemma 4.1, matrix in the right-hand side of equation (4.7) is invertible and we obtain

$$-W_{\perp} (W_{\perp}^T (A(\sigma) - \lambda(\sigma)M) W_{\perp})^{-1} W_{\perp}^T \delta A(\sigma) \bar{x}(\sigma) = W_{\perp} \gamma(\sigma). \quad \square$$

It is straightforward to estimate the $\bar{A}$ -norm of $\bar{x}(\sigma)$ appearing in (4.4). Recall that $\bar{x}(\sigma)^T \bar{A} W_{\perp} = 0$ and using spectral equivalence (3.1) gives

$$\|\bar{x}(\sigma)\|_{\bar{A}} \leq \|x(\sigma)\|_{\bar{A}} \leq \frac{1}{\alpha} \|x(\sigma)\|_{A(\sigma)}. \quad (4.8)$$

As $x(\sigma)$ is an eigenvector of the pencil $(A(\sigma), M)$ corresponding to an eigenvalue on $(0, \Lambda)$ and $\|x(\sigma)\|_M = 1$, it follows that

$$\|x(\sigma)\|_{A(\sigma)} \leq \Lambda^{1/2} \quad \text{and thus} \quad \|\bar{x}(\sigma)\|_{\bar{A}} \leq \frac{1}{\alpha} \Lambda^{1/2}. \quad (4.9)$$